



Tour de Maths 2018

Leg 7

Memo

Question 1

Find the smallest natural number which, when divided by 2, 3, 4, 5 and 6 respectively, leaves a remainder of 1 every time.

Answer: 61

LCM of 2, 3, 4, 5 and 6 is 60.

$$60 + 1 = 61$$

Question 2

The sum of two positive integers is 104. What is the difference between the greatest and least possible products of the two numbers?

Answer: 2601

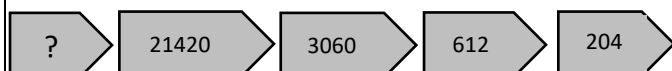
$$52 + 52 = 104 \text{ and } 52 \times 52 = 2704 \text{ (greatest)}$$

$$103 + 1 = 103 \text{ and } 103 \times 1 = 103 \text{ (least)}$$

$$2704 - 103 = 2601$$

Question 3

What is the first number in this pattern?



Answer: 192780

$$612 \div 204 = 3$$

$$3060 \div 612 = 5$$

$$21420 \div 3060 = 7$$

$$\therefore ? = 21420 \times 9$$

$$? = 192780$$

Question 4

In the magic square, the sum of the three numbers in each row, column and diagonal adds up to 30. What number comes in block A?

z	A	$11\frac{2}{3}$
$18\frac{1}{3}$	y	x
		$16\frac{2}{3}$

Answer: A = 15

$$x = 30 - \left(16\frac{2}{3} + 11\frac{2}{3}\right) = 1\frac{2}{3}$$

$$y = 30 - \left(1\frac{2}{3} + 18\frac{1}{3}\right) = 10$$

$$z = 30 - \left(16\frac{2}{3} + 10\right) = 3\frac{1}{3}$$

$$A = 30 - \left(3\frac{1}{3} + 11\frac{2}{3}\right) = 15$$

Question 5

If A , B and C are all positive integers, calculate the value of C if:

$$A \times B = 48, A + B = 19 \text{ and } A + C = 14$$

Answer: 11

Find the factors of 48 that, when added together, are 19. These are 16 and 3. So either A is 16 or 3, or B is 16 or 3.

In order for $A + C$ to be 14, A must be less than 14. So $A = 3$.

$$A + C = 14$$

$$C = 14 - (3)$$

$$C = 11$$

Question 6

One toffee and one chocolate bar cost R9,00. Ten toffees and five chocolate bars cost R47,00. How much does one chocolate bar cost?

**Answer: R8,60**

Let the number of toffees be x and the number of chocolate bars be y .

$$\text{Then } x + y = 9 \text{ ie } x = 9 - y$$

$$\text{And } 10x + 5y = 47$$

Substitute $(9 - y)$ for x in the second equation.

$$\therefore 10(9 - y) + 5y = 47$$

$$90 - 10y + 5y = 47$$

$$-5y = -43$$

$$y = 8,6 \text{ so one chocolate bar costs R8,60}$$

Question 7

A rectangular swimming pool is 3 times as long as it is wide. If it were 5 metres shorter and 3 metres wider, it would be square. What is the perimeter of the swimming pool?

Answer: 32m

The rectangular pool's dimensions are $3x$ by x . If the longest side was 5 metres shorter, it would be $3x - 5$ and if the shortest side was 3 metres wider it would be $x + 3$. If this made the pool square, these new dimensions would be equal to each other.

$$3x - 5 = x + 3$$

$$3x - x = 3 + 5$$

$$2x = 8$$

$$x = 4$$

This means the rectangular pool has a longest side of 12m and a shortest side of 4m. It's perimeter is then:

$$P = 2(12 + 4)m$$

$$P = 2(16)m$$

$$P = 32m$$

Question 8

If $x + y + z = 28$, $x - y - z = 10$ and $z = 2$, then find the value of $2xyz$.

Answer: 532

Substitute 2 for z in both equations.

$$1) \quad x + y + 2 = 28$$

$$x + y = 26$$

$$2) \quad x - y - 2 = 10$$

$$x - y = 12$$

$$\text{and } x = y + 12$$

Substitute $y + 12$ for x in the first equation:

$$y + 12 + y = 26$$

$$2y = 14$$

$$y = 7$$

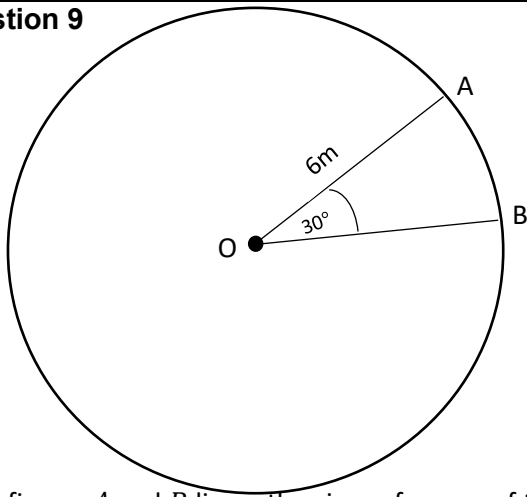
Substitute 7 for y in any equation:

$$x + 7 = 26$$

$$x = 19$$

$$2xyz = 2 \times 19 \times 7 \times 2$$

$$2xyz = 532$$

Question 9

In the figure, A and B lie on the circumference of the circle centred at O . OA is $6m$ long, and the measurement of $\angle AOB$ is 30° . How long is the minor arc AB in terms of π ?

Answer: π

There are 360° in a circle, so this sector AOB is $\frac{30^\circ}{360^\circ}$ which is $\frac{1}{12}$ of the area of the circle, and the arc AB is $\frac{1}{12}$ of the circumference.

$$\begin{aligned}\text{Circumference} &= 2\pi r \\ &= 2\pi(6) \\ &= 12\pi\end{aligned}$$

$$\begin{aligned}\text{Arc } AB &= \frac{1}{12} \times 12\pi \\ &= \pi\end{aligned}$$

Question 11

294 digits are used to number the pages of a book. How many pages does the book have?

Answer: 134

Pages 1-9 have 1 digit each, so 9 digits are used.

Pages 10-99 have 2 digits each which is 90×2 which means another 180 digits.

Up to page 99, 189 digits would have been used.

The number of digits available are:
 $294 - 189 = 105$

All subsequent pages will have 3 digits.
So $105 \div 3 = 35$ more pages.

$$99 + 35 = 134 \text{ pages}$$

Question 10

If two typists can type two pages in two minutes, how many typists will it take to type 18 pages in six minutes?

Answer: 6

If 2 typists can type 2 pages in 2 mins,
then 1 typist can type 1 page in 2 mins

By this reasoning:

x typists can type x pages in 2 mins

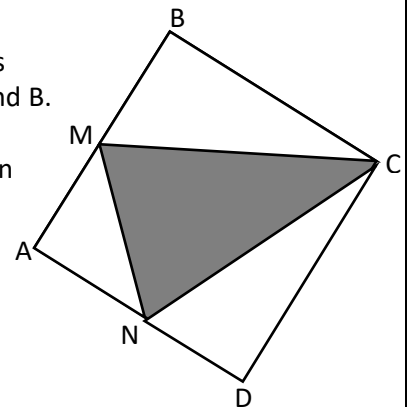
So 18 typists can type 18 pages in 2 mins

To type 18 pages in 6 mins, only a third of these typists are needed, so:

$$18 \div 3 = 6 \text{ typists are needed}$$

Question 12

ABCD is a square. M is halfway between A and B. N is halfway between A and D. What fraction is the shaded part of the square?



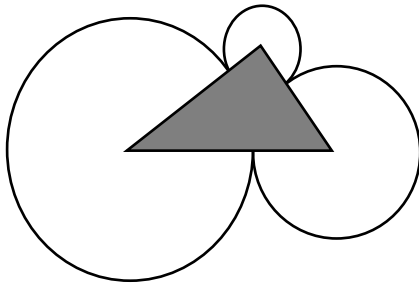
Answer: $\frac{3}{8}$

$\triangle AMN$ is $\frac{1}{8}$ of the area of the large square.

$\triangle CDN$ and $\triangle BCM$ are both $\frac{1}{4}$ of the area of the large square.

$$\begin{aligned}\therefore \text{the shaded area} &= 1 - \left(\frac{1}{8} + \frac{1}{4} + \frac{1}{4}\right) \\ &= 1 - \frac{5}{8} \\ &= \frac{3}{8}\end{aligned}$$

Question 13 The price of a new pair of jeans is decreased by 20%. This new price is then increased by 30%. What percentage is the final price more than the original price? Answer: 4% Let the price of the jeans be x. Initially the price will decrease to 80% of x, so $0,8x$. This new price is increased by 30%, so it will be $1,3 \times 0,8x$ which is $1,04x$. This means the final price is now 4% more than the original price.	Question 14 Today is Thursday, 25 October, 2018. In which year will the next October 25th fall on a Thursday? Answer: 2029 There are 365 days in a year, which is 52 weeks and 1 day. This means any date will be 1 day of the week after it is this year. A leap year will be 2 days later, since 25 October falls after the end of Feb. In 2019 it will be on a Fri. In 2020 there is a leap year so it will fall on a Sun. In 2021 it will be on a Mon. In 2022 it will be on a Tues. In 2023 it will be on a Wed. In 2024 there is a leap year so it will fall on a Fri. In 2025 it will be on a Sat. In 2026 it will be on a Sun. In 2027 it will be on a Mon. In 2028 there is a leap year so it will fall on a Wed. In 2029 it will be on a Thurs.
Question 15 Bob cut a sheet of paper into 8 pieces. Then he took one piece and cut it again into 8 pieces. He then repeated this four more times. How many pieces of paper did he have after the last cutting? Answer: 43 After the 1st cut he would have 8 pieces. Each time he will start with 1 less piece of paper than before but add 8 new pieces. After the 2nd cut he would have $7 + 8 = 15$ After the 3rd cut he would have $14 + 8 = 22$ After the 4th cut he would have $21 + 8 = 29$ After the 5th cut he would have $28 + 8 = 36$ After the 6th cut he would have $35 + 8 = 43$ He would have 43 pieces after he repeated the process 6 times.	Question 16 This morning at 6:00, my watch was 3 minutes slow, but at 13:00 it was 7 minutes fast! At what time was it exactly correct?  Answer: 08:06 The watch caught up 10 minutes in 7 hours. It caught up 1 minute after $\frac{7}{10} \times 60$ minutes, so it caught up 1 minute after 42 minutes. This means it would have caught up the 3 minutes after 3×42 minutes, which is 126 minutes. This is 2 hours and 6 minutes after 6: 00, So it will be at 08: 06.

Question 17

The vertices of the shaded triangle are at the midpoints of each circle. The dimensions of the sides of the triangle are 58, 44 and 36 units. Find the radius of the largest circle.

Answer: 33 units

Let the radius of the largest circle be x .

Let the radius of the medium circle be y .

Let the radius of the largest circle be z .

$$x + y = 58 \quad \text{and therefore } y = 58 - x$$

$$x + z = 44 \quad \text{and therefore } z = 44 - x$$

$$y + z = 36$$

$$\text{so } (58 - x) + (44 - x) = 36$$

$$(102 - 2x) = 36$$

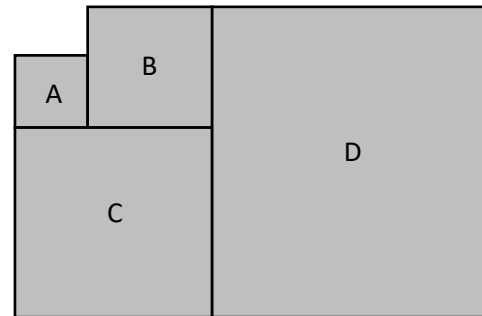
$$-2x = 36 - 102$$

$$x = 33$$

The radius of the largest circle is 33 units.

Question 18

The perimeter of Square A is 120cm and the area of Square B is $0,64\text{m}^2$. Figures C and D are also squares. What is the area of the entire shaded region in m^2 ?



Answer: $2,67\text{m}^2$

If perimeter of A = 120cm , length of side A = 30cm ,
 $30\text{cm} = 0,3\text{m}$. Area of A = $0,09\text{m}^2$.

If area of B = $0,64\text{m}^2$, then length of side B = $0,8\text{m}$.

Length of side C = side length A + side length B

$$= 0,3\text{m} + 0,8\text{m}$$

$$= 1,1\text{m}$$

Area of C = $1,21\text{m}^2$

Length of Side D = side length B + side length C

$$= 0,8\text{m} + 1,1\text{m}$$

$$= 1,9\text{m}$$

Area of D = $3,61\text{m}^2$

Total Shaded Area = Area A + B + C + D

$$= 0,09 + 0,64 + 1,21 + 3,61$$

$$= 5,55\text{m}^2$$

Or Total Shaded Area

= Area of larger rectangle – unshaded area

$$= [(1,1 + 1,9) \times 1,9] - (0,5 \times 0,3)$$

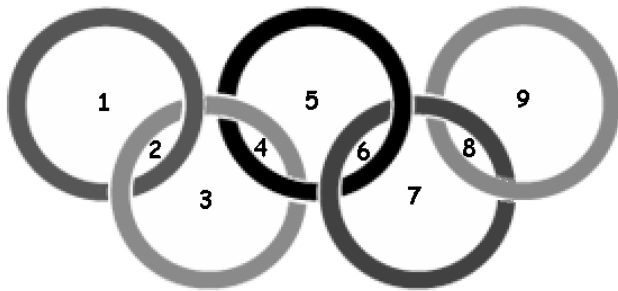
$$= (3,0 \times 1,9) - 0,15$$

$$= 5,7 - 0,06$$

$$= 5,55\text{m}^2$$

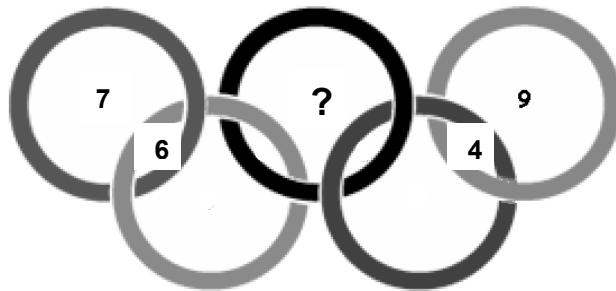
Question 19

Rearrange the digits in the rings so that the sum of digits inside each ring adds up to a total of 13.
What number will appear in the top centre ring?
Hint: Put the 7 in the top left circle, and put the 9 in the top right circle.



Answer: 3

The first ring on the top left needs a 6 to make it 13.
The last ring on the top right needs a 4 to make it 13.

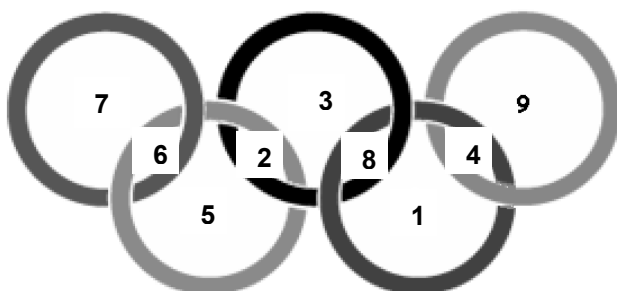


The available numbers are now 1, 2, 3, 5 and 8.

The next 2 numbers in the bottom left ring need to add up to 7, and the only possible combination is 2 and 5.

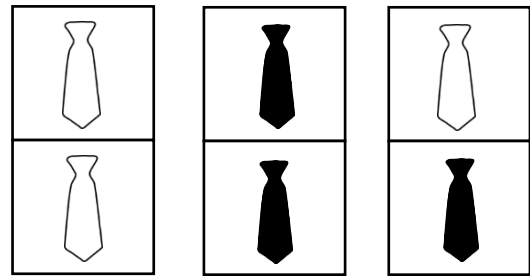
The next 2 numbers in the bottom right ring need to add up to 9, and the only possible combination is 1 and 8.

This leaves only the 3 for the centre of the top middle ring.



Question 20

There are three boxes, each with two compartments. Each compartment holds either a black tie or a white tie. The ties are placed in compartments and boxes exactly as depicted below.



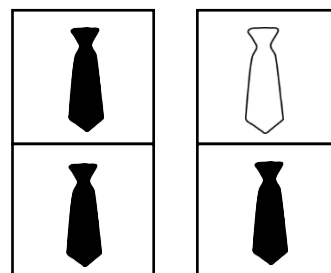
You choose a box at random, then open a compartment at random. If the tie is black, what is the probability that the other tie in the box is also black?

Answer: $\frac{2}{3}$

By choosing the first box at random, you could choose any 1 of the 6 ties in the 6 compartments.

If the tie you chose is black, it rules out the first box that only contains white ties.

It means you have only 4 options, and 1 of those options is definitely black.



No matter which black tie is chosen, it will always leave the option of 2 black ties and 1 white tie.

Therefore, after a black tie is selected first, the remaining probability of getting another black tie is 2 out of 3.